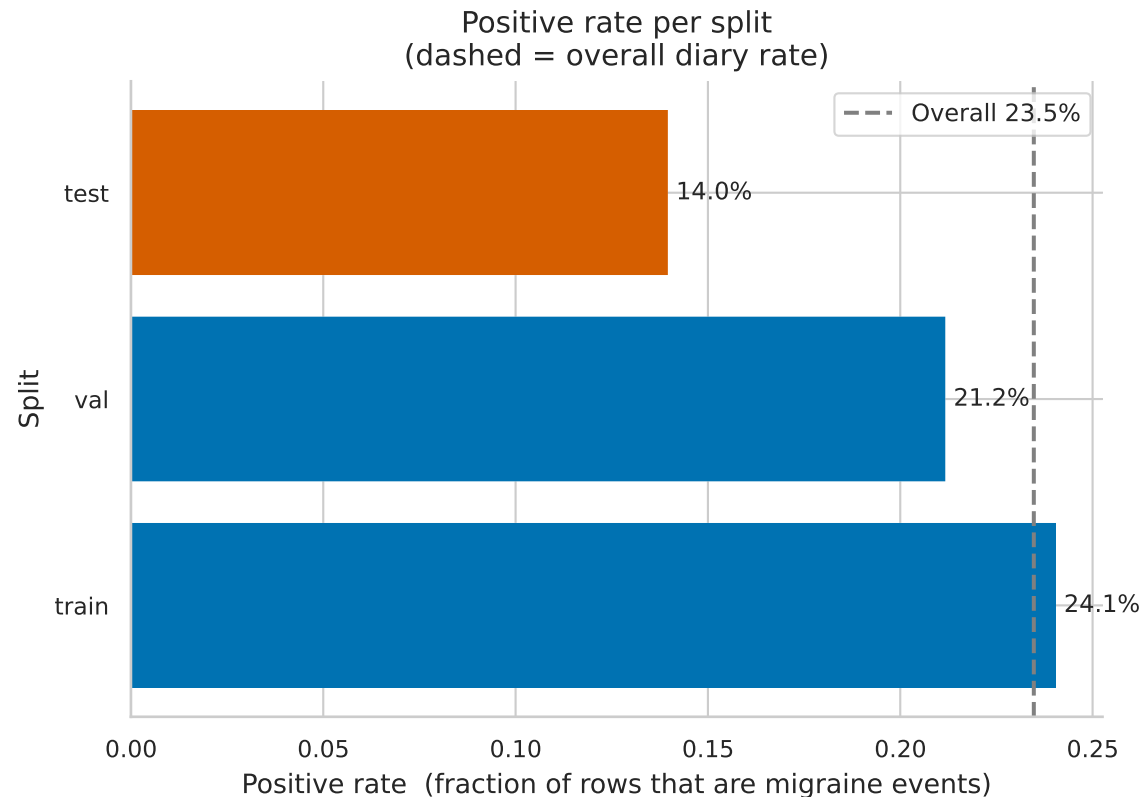
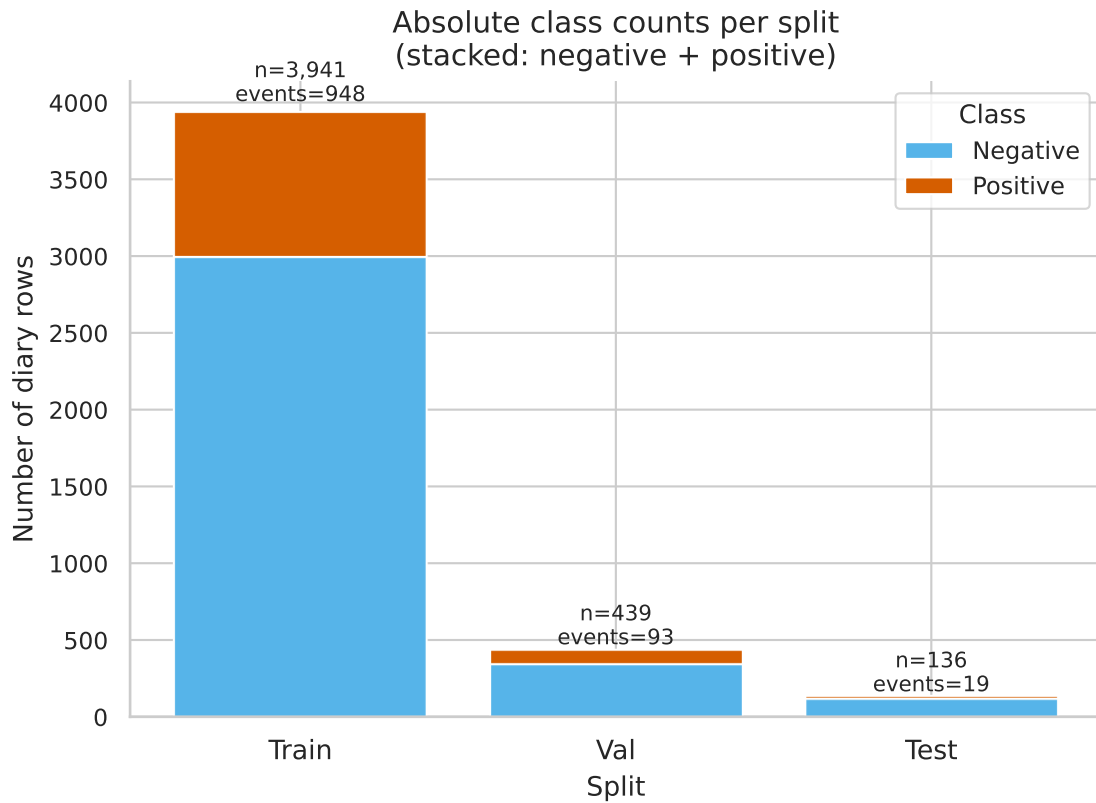


Class Balance Overview — headache
(green: within 5pp of overall; yellow: 5-10pp; red: >10pp)

Package (ratio / strategy)	Train Rate	Val Rate	Test Rate	Train Events	Val Events	Test Events
70_15_15/chrono	24.1%	21.2%	14.0%	948	93	19
70_15_15/patient	23.5%	26.3%	20.7%	759	162	139
70_15_15/stratified	23.5%	23.5%	23.5%	742	159	159
80_20/chrono	24.0%	—	13.7%	1028	—	32
80_20/patient	23.4%	—	23.8%	846	—	214
80_20/stratified	23.5%	—	23.5%	848	—	212
70_30/chrono	24.1%	—	19.5%	948	—	112
70_30/patient	23.5%	—	23.4%	759	—	301
70_30/stratified	23.5%	—	23.5%	742	—	318

Overall diary rate: 23.47%

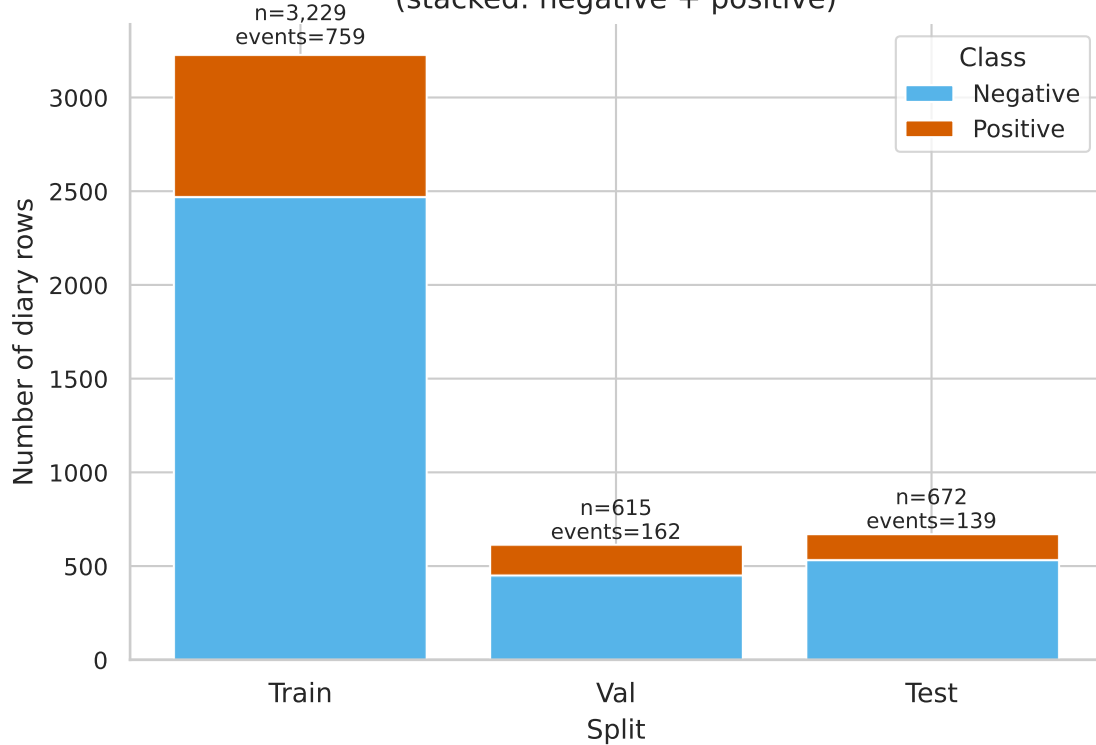
70_15_15/chrono — headache



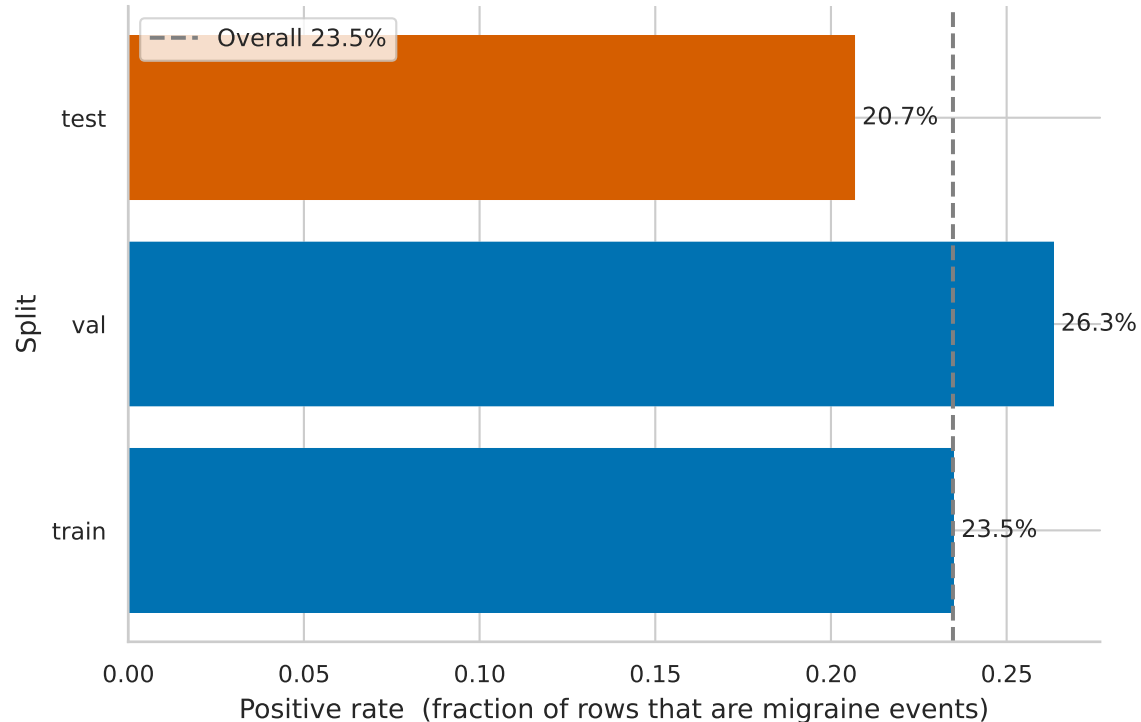
train: imbalance ratio 3.2:1 ($\Delta=+0.6\%$) | val: imbalance ratio 3.7:1 ($\Delta=-2.3\%$) | test: imbalance ratio 6.2:1 ($\Delta=-9.5\%$)

70_15_15/patient — headache

Absolute class counts per split
(stacked: negative + positive)

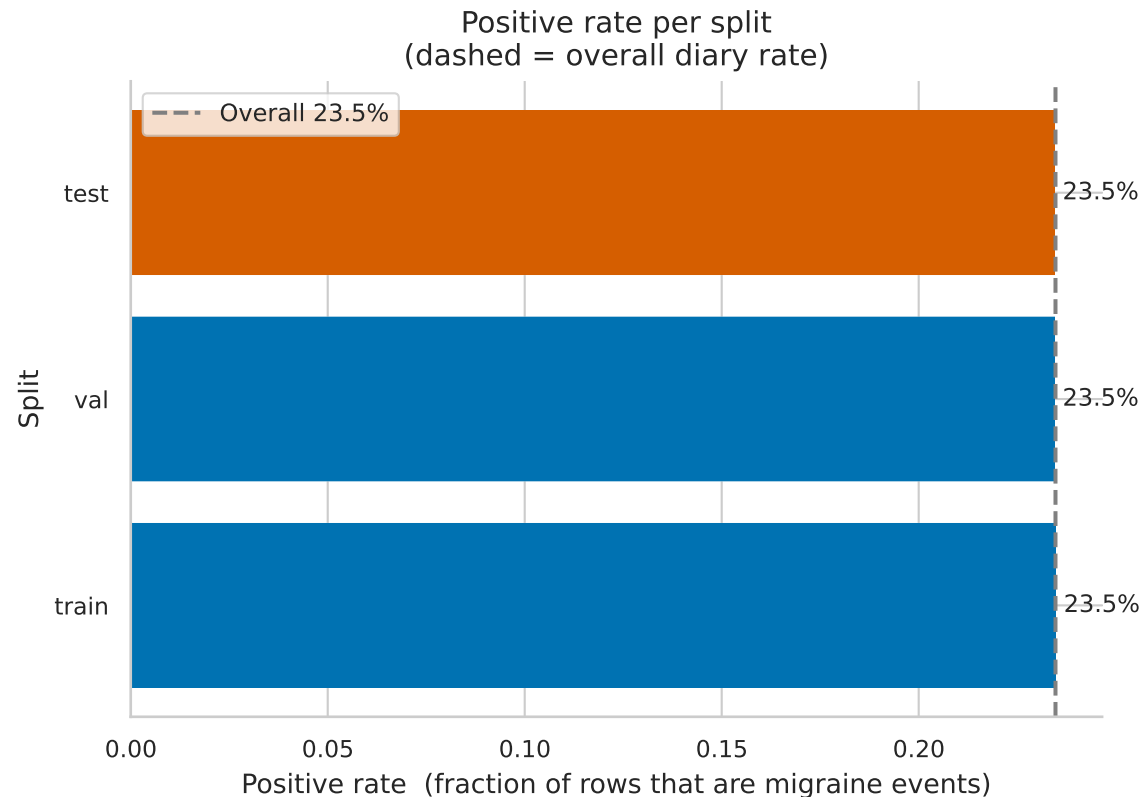
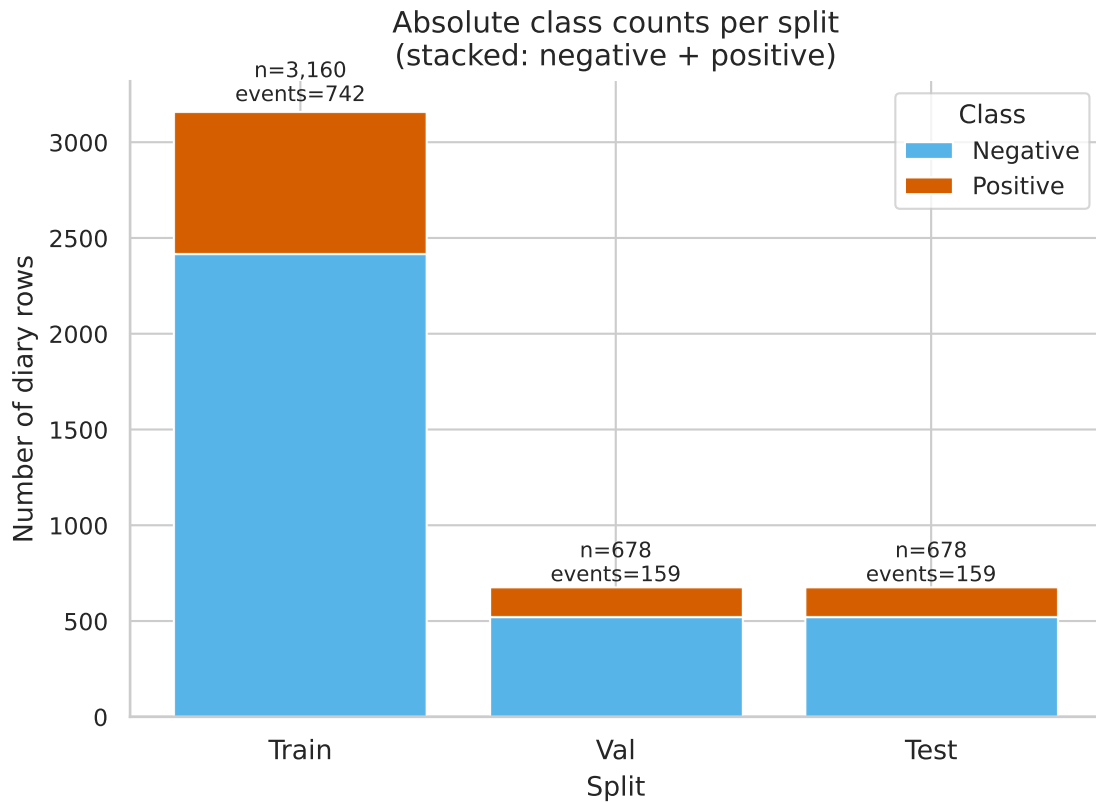


Positive rate per split
(dashed = overall diary rate)



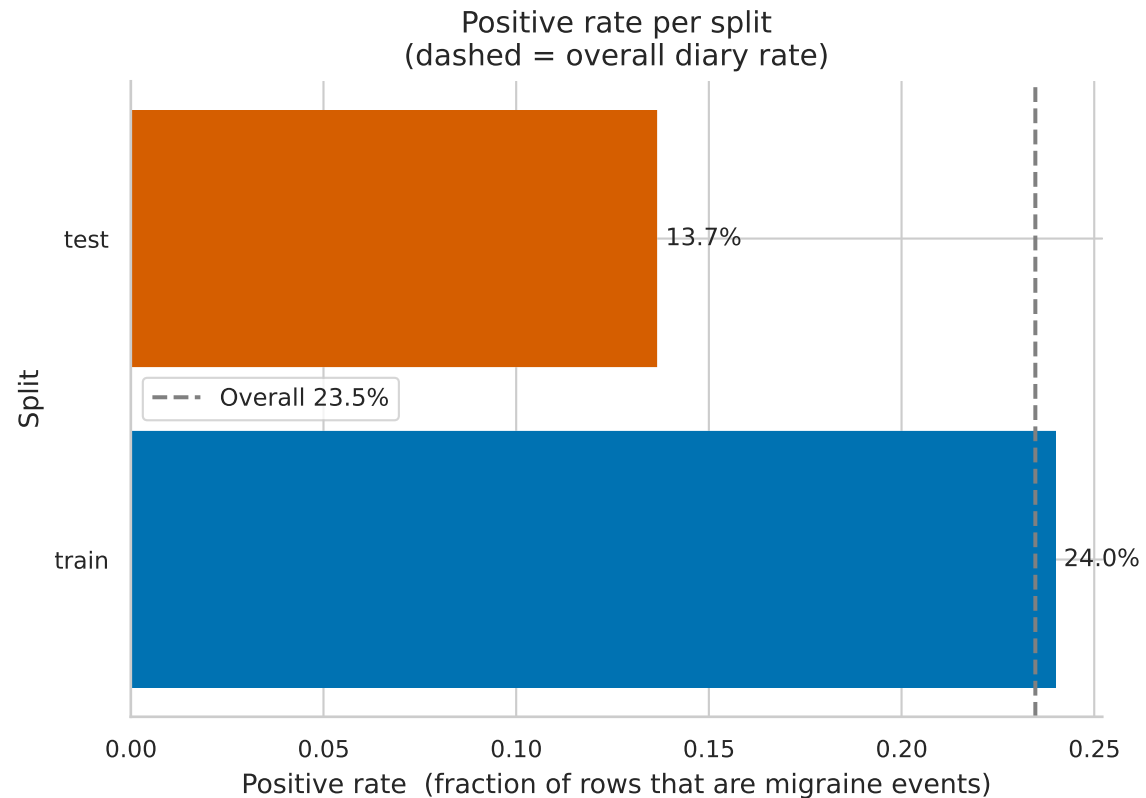
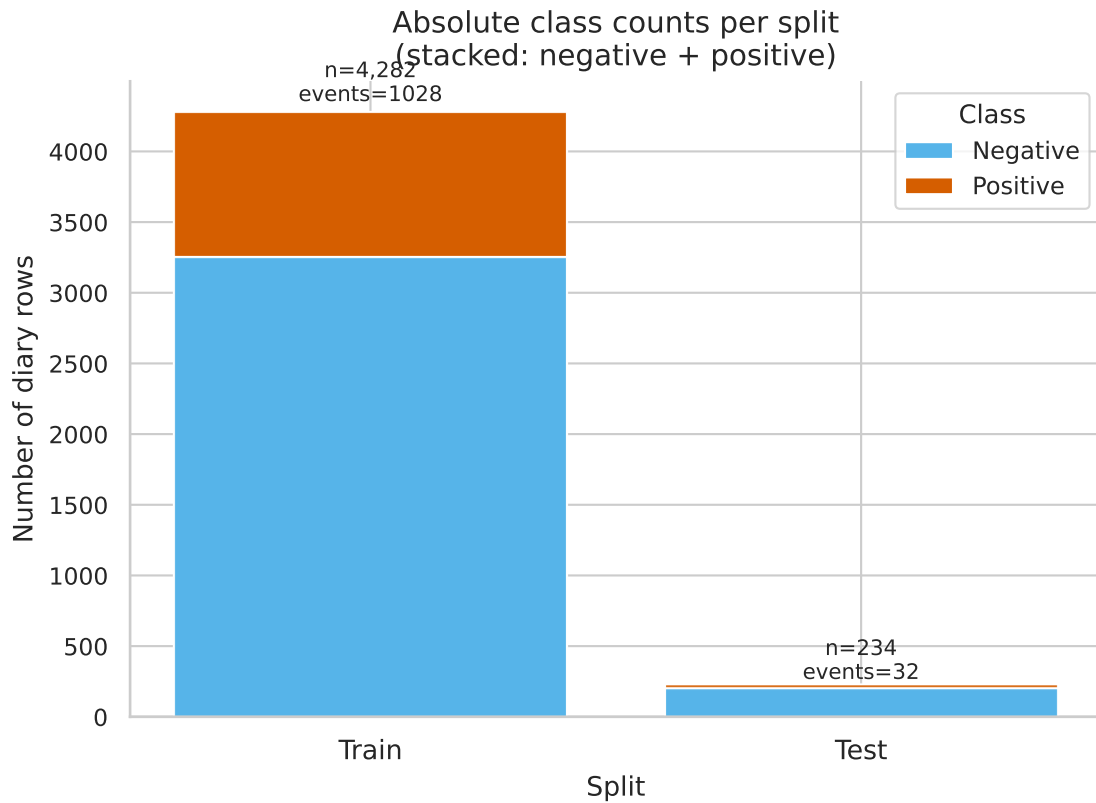
train: imbalance ratio 3.3:1 ($\Delta=+0.0\%$) | val: imbalance ratio 2.8:1 ($\Delta=+2.9\%$) | test: imbalance ratio 3.8:1 ($\Delta=-2.8\%$)

70_15_15/stratified — headache



train: imbalance ratio 3.3:1 ($\Delta=+0.0\%$) | val: imbalance ratio 3.3:1 ($\Delta=-0.0\%$) | test: imbalance ratio 3.3:1 ($\Delta=-0.0\%$)

80_20/chrono — headache

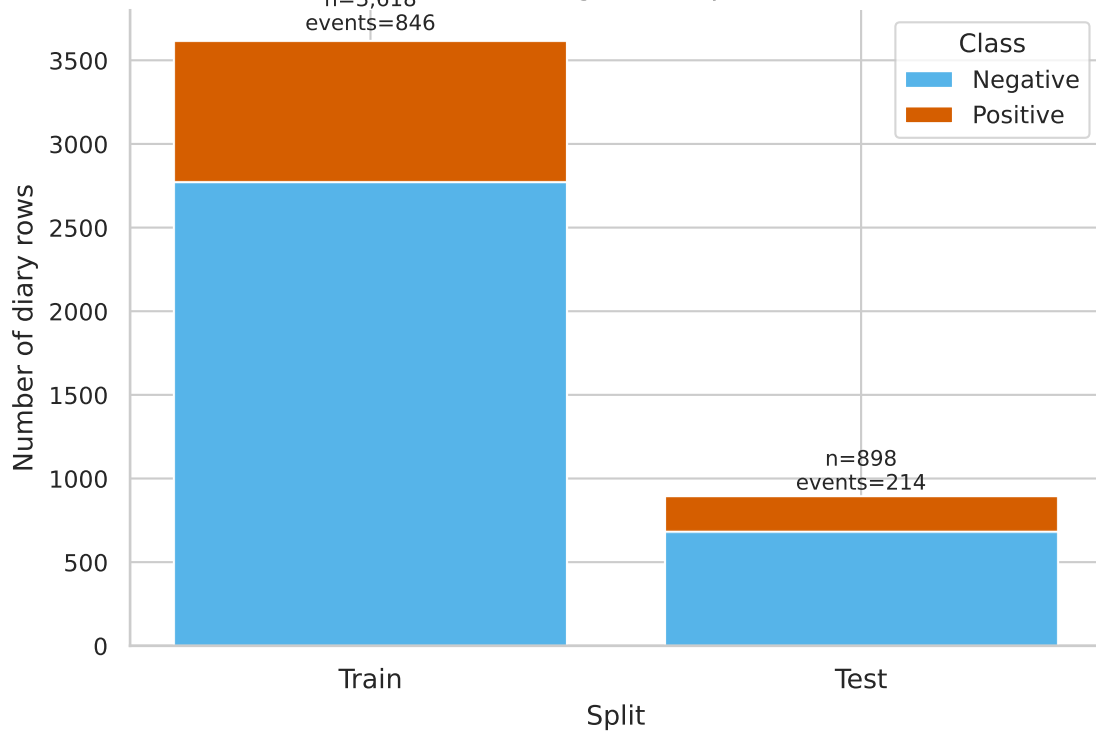


train: imbalance ratio 3.2:1 ($\Delta=+0.5\%$) | test: imbalance ratio 6.3:1 ($\Delta=-9.8\%$)

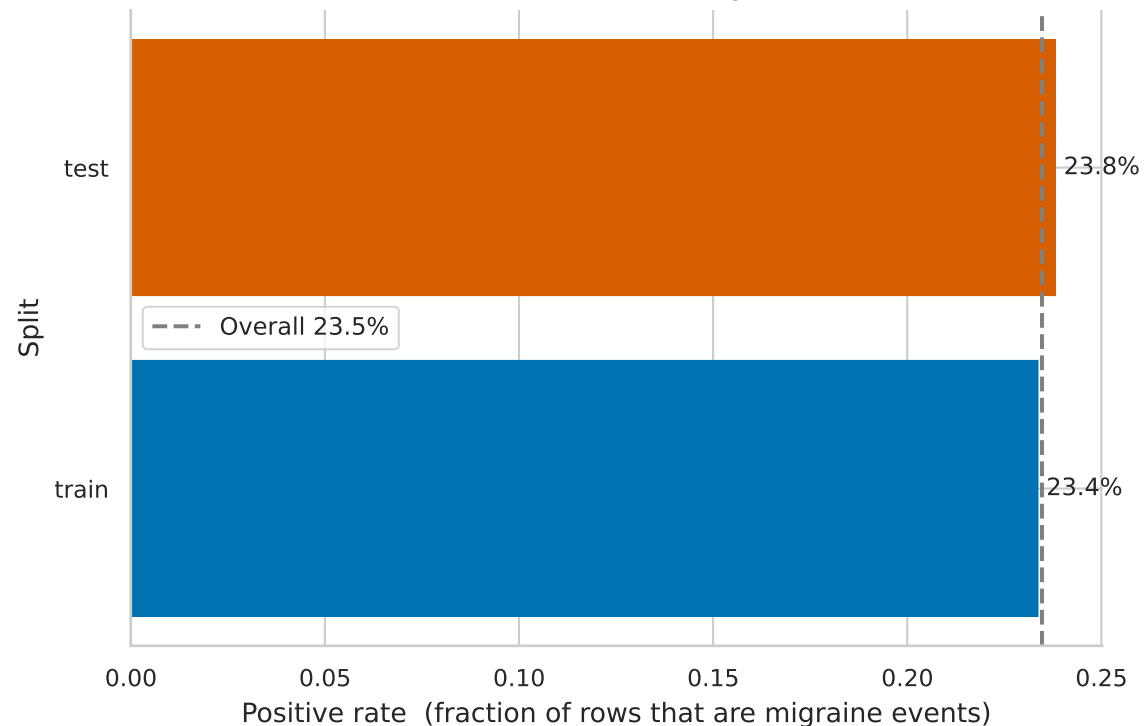
80_20/patient — headache

Absolute class counts per split
(stacked: negative + positive)

n=3,618
events=846

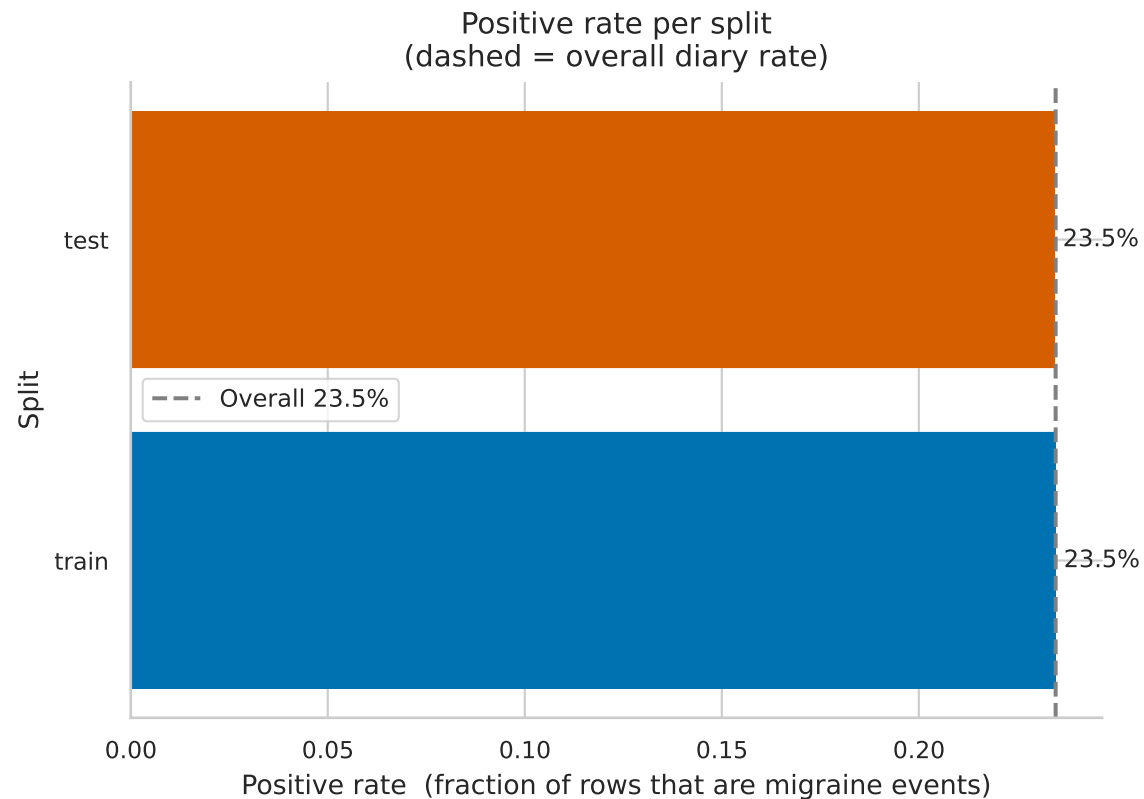
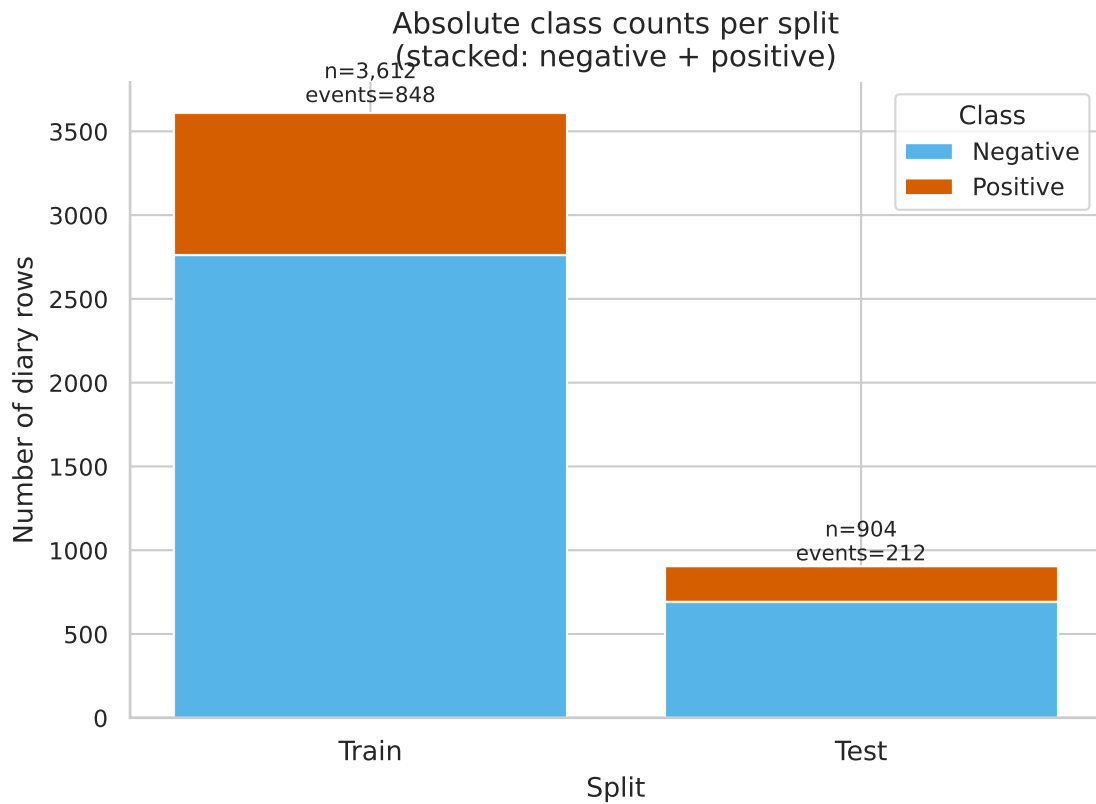


Positive rate per split
(dashed = overall diary rate)



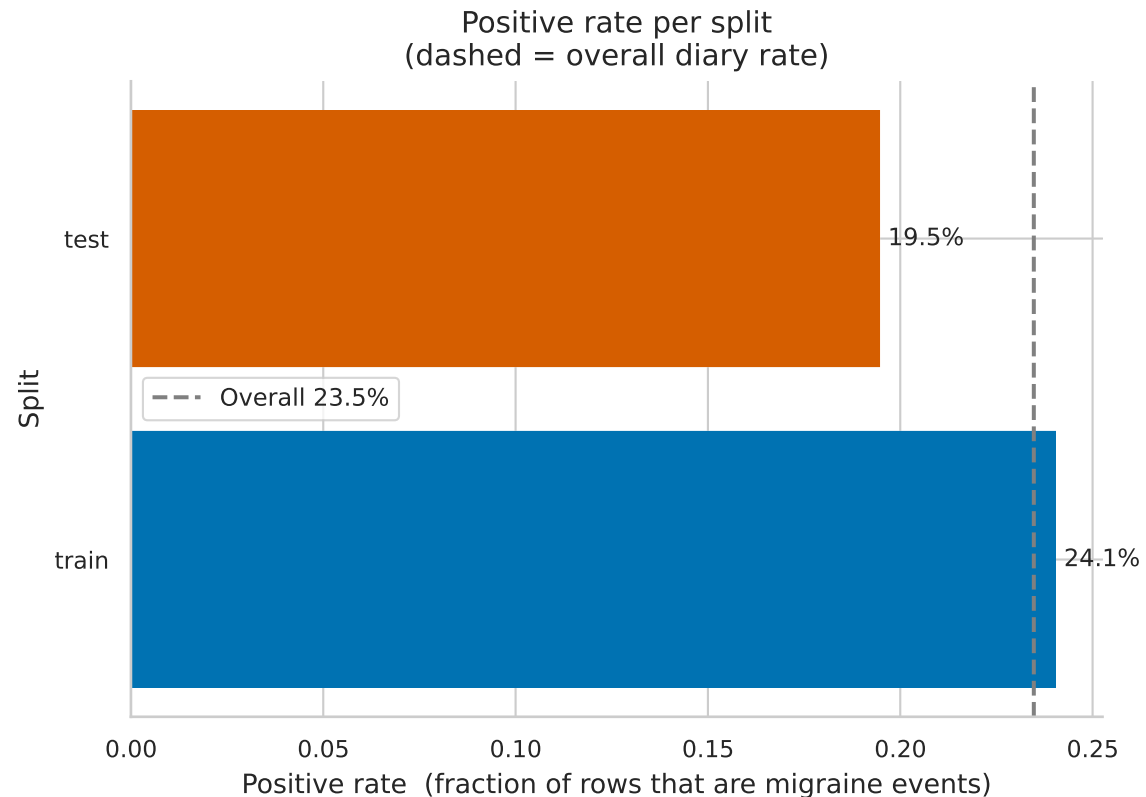
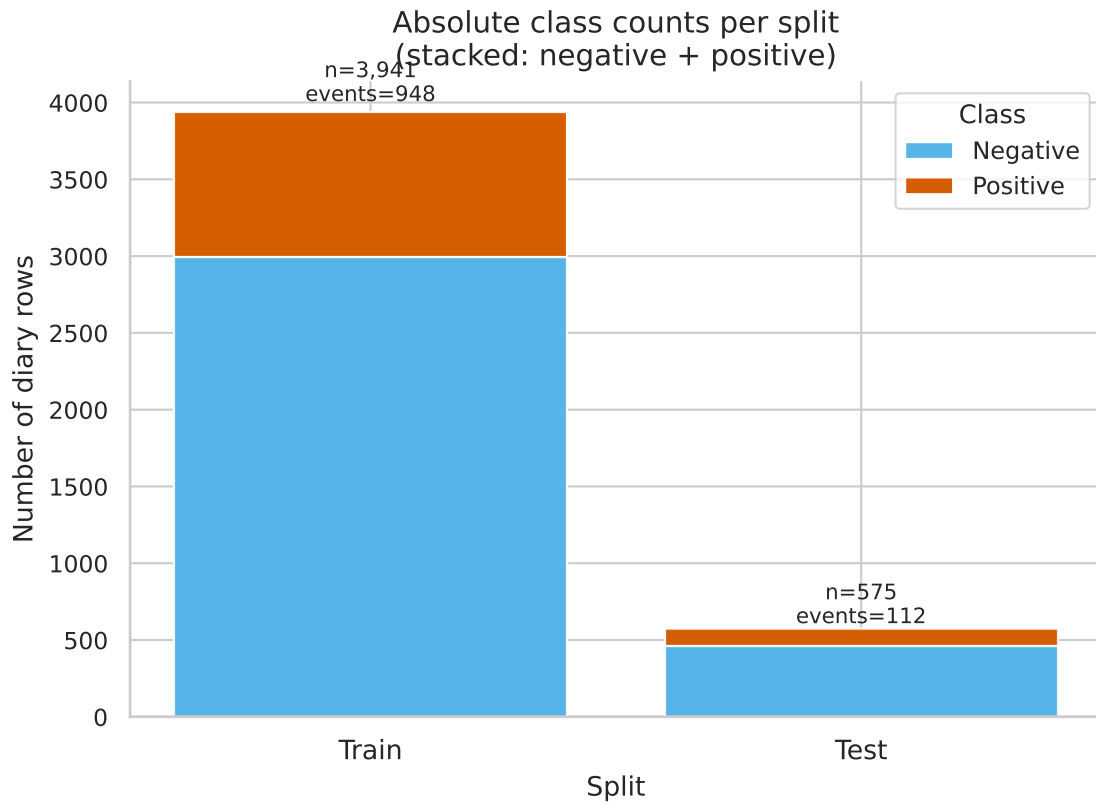
train: imbalance ratio 3.3:1 ($\Delta=-0.1\%$) | test: imbalance ratio 3.2:1 ($\Delta=+0.4\%$)

80_20/stratified — headache



train: imbalance ratio 3.3:1 ($\Delta=+0.0\%$) | test: imbalance ratio 3.3:1 ($\Delta=-0.0\%$)

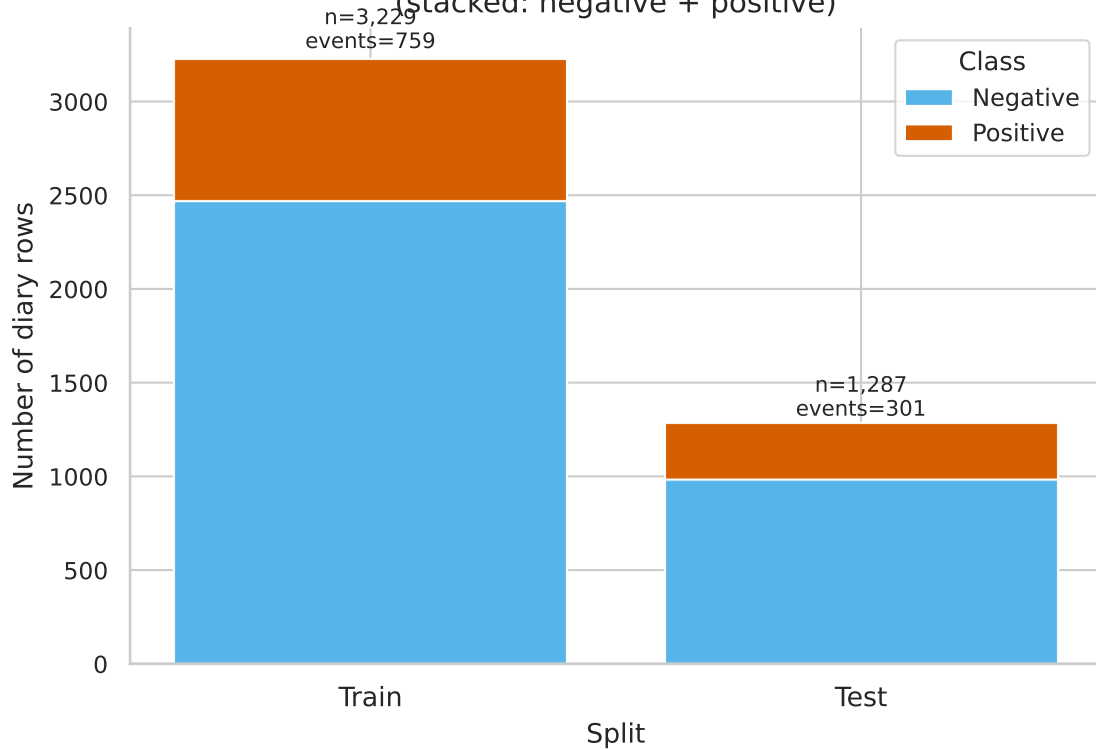
70_30/chrono — headache



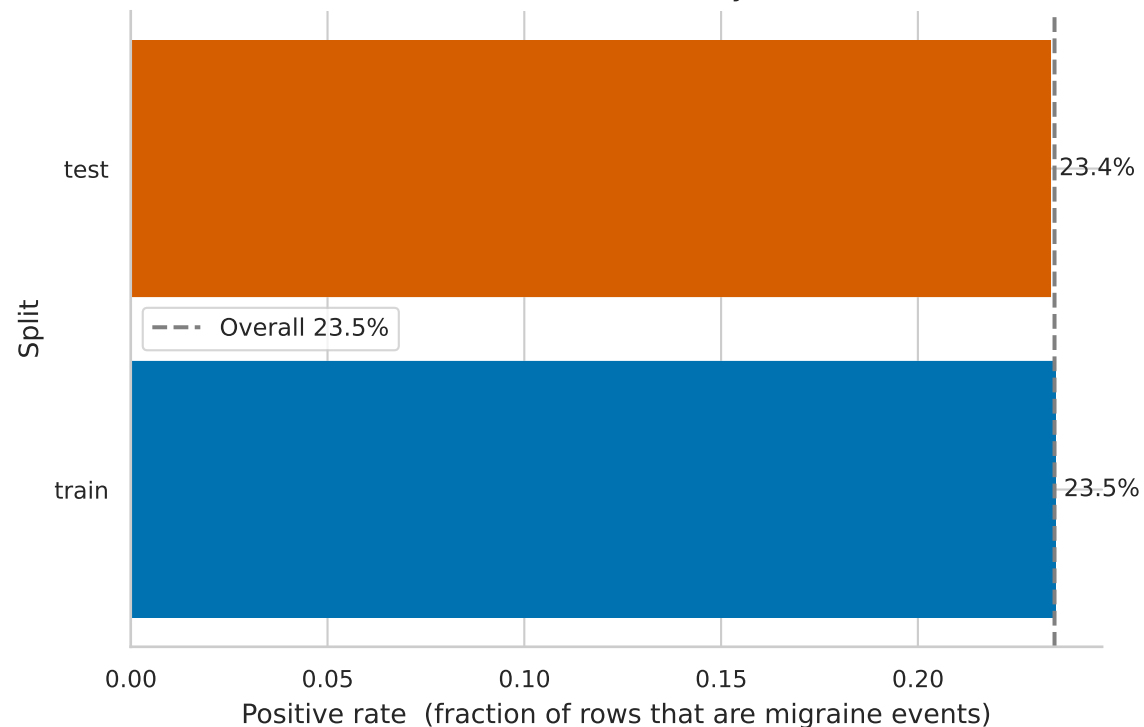
train: imbalance ratio 3.2:1 ($\Delta=+0.6\%$) | test: imbalance ratio 4.1:1 ($\Delta=-4.0\%$)

70_30/patient — headache

Absolute class counts per split
(stacked: negative + positive)

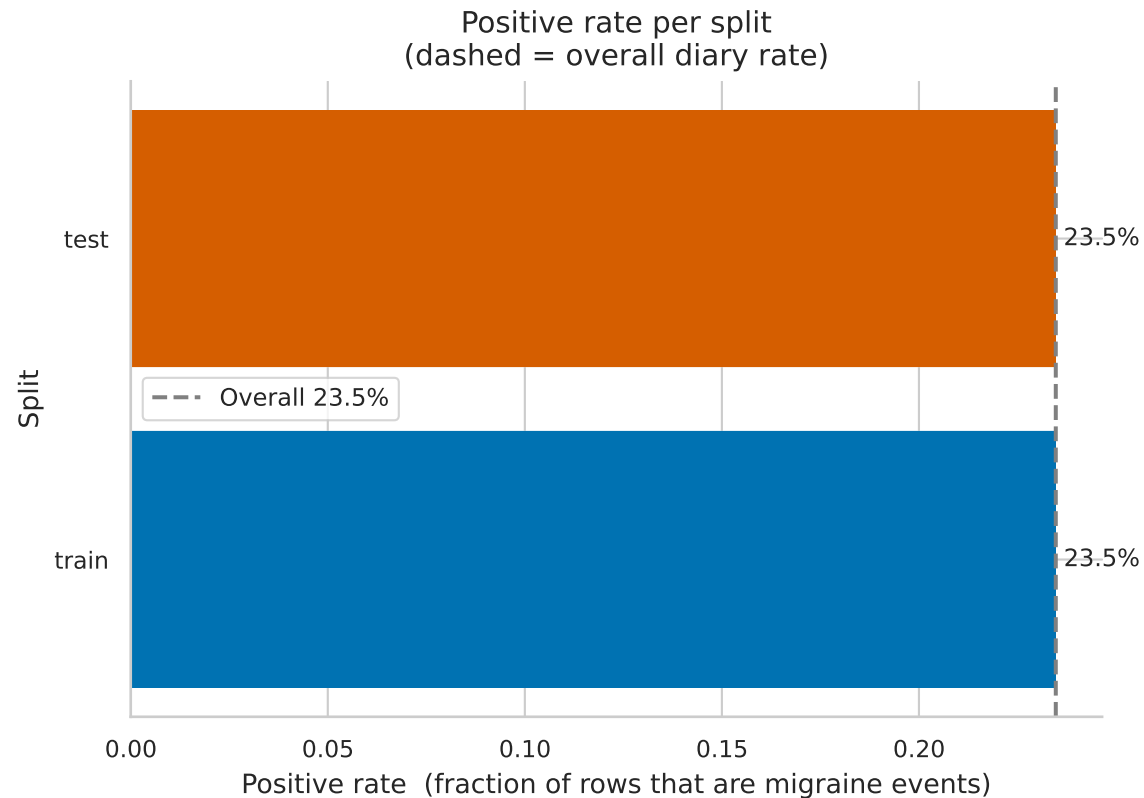
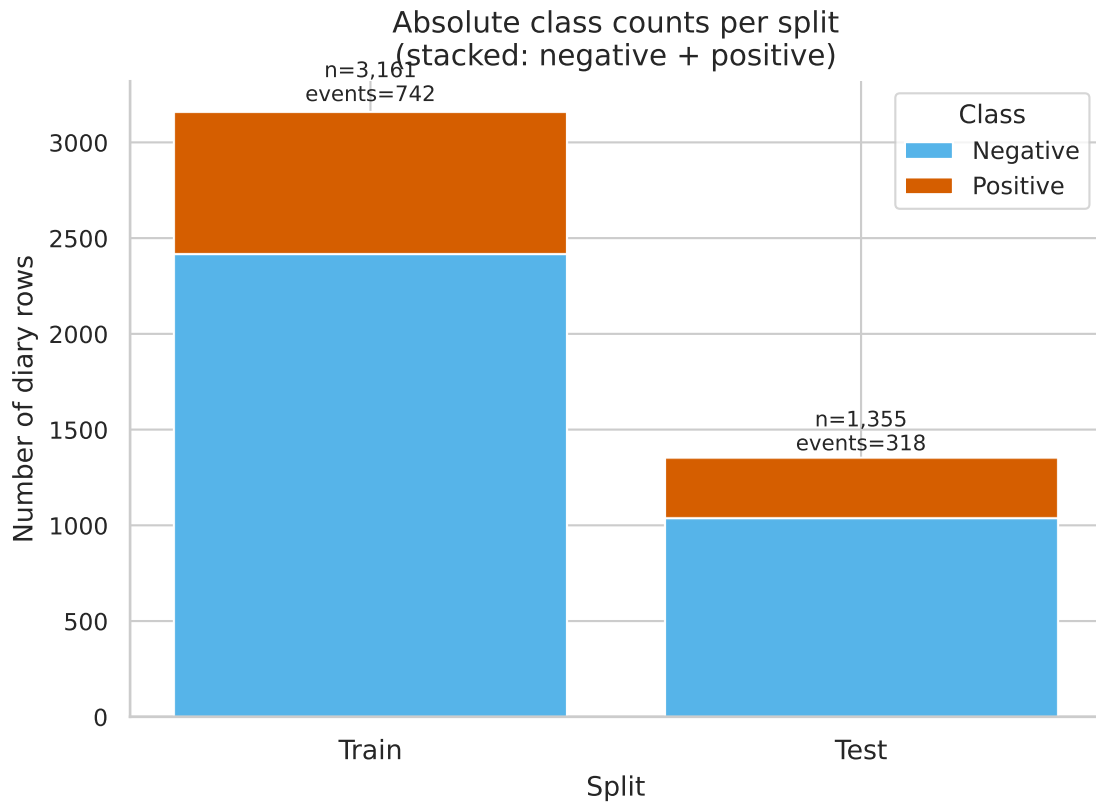


Positive rate per split
(dashed = overall diary rate)



train: imbalance ratio 3.3:1 ($\Delta=+0.0\%$) | test: imbalance ratio 3.3:1 ($\Delta=-0.1\%$)

70_30/stratified — headache



train: imbalance ratio 3.3:1 ($\Delta=+0.0\%$) | test: imbalance ratio 3.3:1 ($\Delta=-0.0\%$)

Cross-package Test Positive Rate — headache

Train vs test positive rate across all 9 packages
Red annotation = train-test delta > 5 percentage points

